

# SURENDHAR E R

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Erode, Tamil Nadu, India

## Summary

Computer Science student and **Frontend Developer and Generative AI Engineer** experienced in building AI-powered applications, RAG pipelines, and scalable systems. Skilled in Python, JavaScript, Java, React, REST APIs, SQL, TensorFlow, and PyTorch. Strong foundation in machine learning, data systems, and modern UI engineering with experience delivering AI-based solutions.

## Experience

### NRIT UpSkills Software Solutions Pvt Ltd

Dec 2025 – May 2026 *Generative AI Engineer*

- Developed AI-powered LMS features using Generative AI and RAG pipelines.
- Built automated question-generation systems from documents and transcripts.
- Designed multi-agent AI workflows for educational content generation.
- Implemented AI-based career guidance and psychometric analysis modules.
- Optimized frontend performance and scalable UI architecture using React.

### Prayana Electric

Sep 2025 – Nov 2025 *UI/UX Designer Intern*

- Designed responsive web and mobile interfaces using user-centered design.
- Created wireframes and prototypes using Figma.

### NOVI TECH R&D Pvt Ltd

Apr 2025 – May 2025 *UI/UX Intern*

- Conducted user research and requirement analysis.
- Designed high-fidelity UI mockups and interactive prototypes.

## Education

### Vellore Institute of Technology – Andhra Pradesh

2022 – Present

*M.Tech (Integrated) Computer Science Engineering*

*CGPA: 7.86*

*(Virtusa Collaboration)*

## Technical Skills

**Programming Languages:** Python, Java, JavaScript **Frontend:** React.js, HTML, Tailwind CSS **Backend:** FastAPI, REST APIs **Databases:** MySQL, MongoDB, SQL **Artificial Intelligence:** TensorFlow, PyTorch, scikit-learn, RAG Pipelines, LLM Integration, Embeddings, Prompt Engineering **Tools:** GitHub, n8n, Jupyter Notebook, Botpress, Figma **Concepts:** Data Structures, DBMS, SDLC

## Projects

### AI-driven UI Automation using NLU

- Developed AI system for generating UI layouts using natural language input.
- Improved structured UI generation and accessibility.

### AI Agriculture Chatbot

- Built chatbot for agriculture queries with improved intent recognition.

### Plant Disease Detection

- Developed deep learning model for image-based plant disease classification.

### HeartCare – Patient Management System

- Built healthcare system with dashboards, appointments, and patient records.

## Research Work

### Aegis Clinical Decision Support System (CDSS)

Sep 2025 – Dec 2025

- Developed a full-stack diagnostic system using FastAPI and React for end-to-end clinical analysis.
- Trained an XGBoost model on 3,600+ records achieving ~82% accuracy.
- Implemented SHAP explainability for 9+ clinical features to improve model transparency.
- Built a RAG pipeline using FAISS with 384-dimensional embeddings, retrieving 12+ medical articles per query.
- Designed a 3-tier LLM system (Gemini, LLaMA, fallback) ensuring 100% response reliability.
- Integrated drug recommendation APIs and OTP-based authentication for secure patient access.

### AgroChain Food Supply Chain and MicroFinance Using Blockchain

2026 – Present

- Integrating blockchain with AI for agricultural transparency and financing.
- Exploring Generative AI and RAG pipelines in decision support systems.

## Certificates

Oracle Generative AI Certification 2025 Generative AI – HCL GUVI UX Certification – Georgia Tech University Basics of Blockchain – University at Buffalo